Lecture 33

First-order differential equations

Uses.

Summary 5: Continuity and one-sided limits are defined by error bounds.
Summary 6: A continuous scalar function on an interval takes every intermediate value.
Summary 8: Product, chain, inverse and elementary derivative rules hold on their stated domains.
Summary 9: A derivative with strict constant sign gives strict monotonicity; a zero derivative gives a constant on an interval.
Summary 12: Euclidean distance and the norm measure errors in the plane.
Summary 16: Complex functions of a real parameter differentiate componentwise. The exponential never vanishes and $(e^{\lambda t})' = \lambda e^{\lambda t}$ for $\lambda \in \mathbb{C}$.
Summary 23: Primitives on an interval differ by a constant. Substitution and integration by parts evaluate primitives.
Summary 26: The fundamental theorem differentiates the integral of a continuous function and evaluates definite integrals by primitives.

Objectives. *A differential equation prescribes a derivative using time and the unknown quantity itself. An initial value selects a solution, when one exists uniquely. Separation and an integrating factor construct solutions for two classes of equations. Their proofs also identify the intervals on which the formulas solve the original problem.*

By the end of this lecture you will be able to:

- State the local continuity and Lipschitz hypotheses of the cited existence and uniqueness theorem.
- Solve a separable equation on intervals where division is valid, retaining its equilibrium solutions.
- Prove and apply the integrating-factor formula for a scalar linear initial-value problem with continuous coefficients.
- Prove uniqueness of $z' = \lambda z$ for a complex-valued function of a real parameter.

33.1 An equation and its initial value

An *ordinary differential equation* relates a function of one real variable to its derivatives. Its *order* is the highest derivative that occurs. A scalar first-order equation in explicit form is

$$y'(t) = F(t, y(t)).$$

The function F is given. The real-valued function y is unknown. A *solution on an open interval* I is a differentiable map $y : I \to \mathbb{R}$ whose graph lies in the domain of F and which satisfies the equation at every $t \in I$. An *initial-value problem* also prescribes $y(t_0) = y_0$, where $t_0 \in I$. The interval belongs to the answer. A formula that becomes undefined at one time cannot be a solution on an interval containing that time.

The value $F(t, y)$ specifies the slope of a solution through the point (t, y) in the time–value plane. A *direction field* draws short segments with those slopes. The segments record the equation rather than an approximation theorem. A proposed curve still needs differentiation and substitution to establish that it is a solution.

For example, $y' = y$ has solutions $y(t) = Ce^t$ for every $C \in \mathbb{R}$: differentiation gives $y' = Ce^t = y$. The initial condition $y(0) = 3$ forces $C = 3$ within this family. That calculation alone does not show that the family contains every solution. A uniqueness argument is needed for that conclusion.

33.2 Local existence and its hypotheses

Continuity of F near a point of the plane means that its value changes by a small amount whenever both coordinates change by a sufficiently small amount. Explicitly, at (t, y), for every $\eta > 0$ there is a $\delta > 0$ such that $|F(s, z) - F(t, y)| < \eta$ whenever the point (s, z) is in the domain and $\sqrt{(s - t)^2 + (z - y)^2} < \delta$. This is the error definition of continuity with the Euclidean distance of Lecture 12. No partial derivatives are required.

A function F is *Lipschitz in the unknown value, uniformly in time*, on a rectangle if one constant $L \geq 0$ satisfies

$$|F(t, y) - F(t, z)| \leq L|y - z|$$

for every time and every pair of values in that rectangle. The same L must work for all those times and pairs.

> **Theorem 33.1** (Local existence and uniqueness, cited). *Let $a, b > 0$ and let F be continuous on the open rectangle $U = (t_0 - a, t_0 + a) \times (y_0 - b, y_0 + b)$. Suppose the displayed Lipschitz inequality holds on U with a finite constant L. Then some open interval I containing t_0 has a C^1 solution of $y' = F(t, y)$, $y(t_0) = y_0$, with graph in U. Any two such solutions agree on some open interval containing t_0. This is the scalar rectangle case of Picard–Lindelöf [C].*

The cited theorem supplies local existence and local uniqueness, not existence for all real time. Its source is Teschl, Theorem 2.2, p. 38. The general constructions below for separable and linear equations prove their stated conclusions independently of this cited result. In one nonlinear example we use the cited theorem separately to justify uniqueness and maximality; that conclusion is marked [C].

For the equation $y' = y^2$, Figure 33.1 shows the direction field and several exact solutions on portions of their domains.

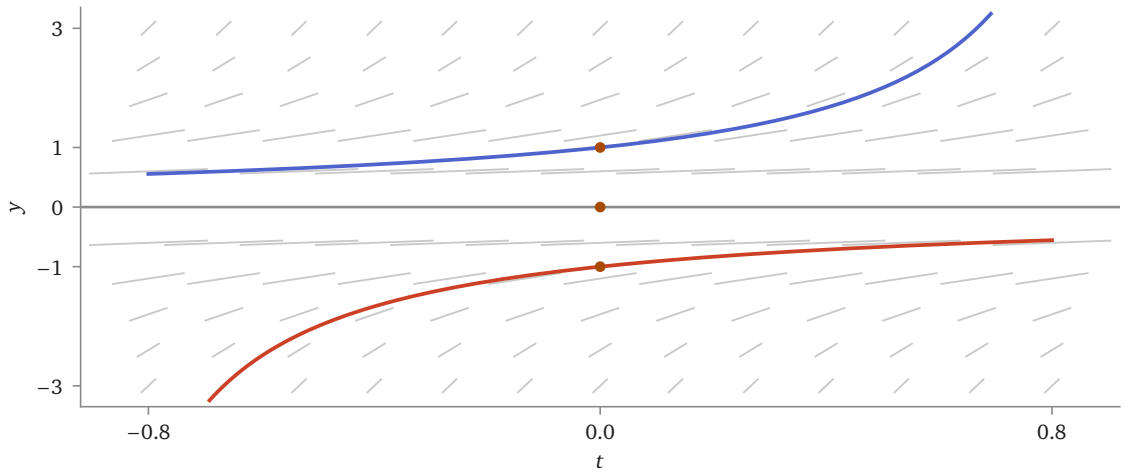


Figure 33.1: Short segments have slope y^2. The curves are exact solutions with initial values -1, 0 and 1 at time zero. Each nonzero curve is drawn only on the side of its pole containing zero.

Example 33.2 (Local uniqueness does not prevent finite-time divergence)**.** Consider $y' = y^2$, $y(0) = 1$. For $|y|, |z| < B$, with $B > 1$, factorisation gives $|y^2 - z^2| = |y + z||y - z| \le 2B|y - z|$. The right side of the equation is continuous and satisfies the local Lipschitz hypothesis. Nevertheless, the solution

$$y(t) = \frac{1}{1 - t} \qquad (-\infty < t < 1)$$

becomes unbounded as $t \to 1^-$. Indeed, its derivative is $1/(1 - t)^2 = y(t)^2$, and $y(0) = 1$. A continuous real-valued extension across 1 would have a finite value there, contrary to this limit. Local existence gives no such extension.

Example 33.3 (Continuity alone does not give uniqueness)**.** Let $F(t, y) = 3|y|^{2/3}$ and prescribe $y(0) = 0$. The right side is continuous, but a finite Lipschitz constant near zero would require $3|y|^{-1/3} \le L$ for all sufficiently small nonzero y. That is impossible. Besides the zero solution, every $c \ge 0$ gives

$$y_c(t) = \begin{cases} 0, & t \le c, \\ (t - c)^3, & t > c. \end{cases}$$

At $t = c$, the derivative from both sides is zero, and the derivative is continuous there. On the right, $y_c' = 3(t - c)^2 = 3|y_c|^{2/3}$. On the left both sides of the equation vanish, including at the joining point. All these C^1 functions are solutions on $\mathbb{R}$ with the same initial value. For $c = 0$ the nonzero branch starts immediately after the initial time. The failure of the Lipschitz hypothesis therefore accompanies actual local nonuniqueness.

33.3 Separation and equilibrium solutions

A *separable equation* has the form

$$y' = A(t)B(y),$$

where A and B are continuous on open intervals I and J, respectively. A constant solution $y(t) = \eta \in J$ is an *equilibrium* when $B(\eta) = 0$. Substitution verifies that solution for every $t \in I$. List these zeros before dividing by $B(y)$.

Theorem 33.4 (Separation where the divisor is nonzero). *Suppose B has no zero on J and let $t_0 \in I$, $y_0 \in J$. Define primitives with fixed base points by*

$$H(y) = \int_{y_0}^{y} \frac{\mathrm{d}u}{B(u)}, \qquad G(t) = \int_{t_0}^{t} A(s)\,\mathrm{d}s.$$

A C^1 function $y : K \to J$, on an open interval $K \subset I$ containing t_0, solves the initial-value problem exactly when $H(y(t)) = G(t)$ for every $t \in K$. There is a unique such solution near t_0, obtained by the local inverse of H.

Proof. If y solves the equation and has the prescribed value, the chain rule and fundamental theorem give

$$\frac{\mathrm{d}}{\mathrm{d}t} H(y(t)) = \frac{y'(t)}{B(y(t))} = A(t) = G'(t).$$

The difference is constant on K. At t_0 both terms are zero, so the constant is zero and $H(y(t)) = G(t)$. Conversely, differentiating that equality yields $y'/B(y) = A(t)$. Multiplication gives the original equation. At t_0 the equality gives $H(y(t_0)) = H(y_0) = 0$. The function B has a constant strict sign on J: otherwise the intermediate value theorem would give a zero. Thus $H' = 1/B$ has a constant strict sign, so H is strictly monotone and the initial value follows by injectivity.

The derivative $H'(y_0) = 1/B(y_0)$ is nonzero and continuous near y_0. The one-variable inverse rule gives a differentiable local inverse of H near $H(y_0) = 0$. Continuity of G, with $G(t_0) = 0$, allows the construction $y(t) = H^{-1}(G(t))$ for all t sufficiently close to t_0. The inverse rule makes this function C^1 and the established converse makes it a solution. Any other local solution with values in that inverse interval has the same expression, proving local uniqueness. ∎

The relation $H(y) = G(t)$ is often written as $\int \mathrm{d}y/B(y) = \int A(t)\,\mathrm{d}t + C$. Its justification is the chain rule above. The notation does not permit division at an equilibrium. If a solution reaches a zero of B, the separated formula applies only on the subintervals where that zero is avoided. The delayed-cubic example shows why a missing zero cannot be repaired merely by adding one arbitrary constant to the divided formula.

Example 33.5 (Decay and a measured half-life). Assume a nonnegative amount y loses material at a rate proportional to its current amount, with constant $k > 0$. The prescribed model is $y' = -ky$ and $y(0) = y_0 \geq 0$. The equilibrium is $y = 0$. On a positive-value interval, separation gives $(\ln y)' = -k$, hence $\ln y = \ln y_0 - kt$ when $y_0 > 0$. The resulting solution, including the equilibrium case, is

$$y(t) = y_0 e^{-kt}.$$

Direct substitution verifies this formula on $\mathbb{R}$; physical future time is $t \geq 0$. Uniqueness for all real initial values will also follow from the linear theorem below. If a half-life is 6 hours, then $e^{-6k} = 1/2$ and $k = (\ln 2)/6$ per hour. For initial amount 80 grams the model becomes

$$y(t) = 80\,2^{-t/6} \text{ grams}.$$

At 9 hours the amount is $20\sqrt{2}$ grams. The first time it reaches 5 grams solves $2^{-t/6} = 1/16$, so it is 24 hours. The amount stays positive at every finite future time and tends to zero as $t \to \infty$.

Example 33.6 (A nonlinear equation and a finite endpoint)**.** Solve $y' = y(y-1)$ with $y(0) = 2$. The constant functions $y = 0$ and $y = 1$ are equilibria. Since the initial value is greater than 1, first work on the value interval $(1, \infty)$, where division by $y(y-1)$ is legal. Partial fractions give

$$\frac{1}{y(y-1)} = \frac{1}{y-1} - \frac{1}{y}, \qquad \ln\left(\frac{y-1}{y}\right) = t + C.$$

The initial value gives $C = -\ln 2$, hence

$$y(t) = \frac{1}{1 - e^t/2}, \qquad t < \ln 2.$$

On this whole interval $y > 1$, so the division used in its derivation is valid. Direct differentiation verifies $y' = y(y-1)$ and $y(0) = 2$. The right side is locally Lipschitz in y: on any bounded value interval, $|y(y-1) - z(z-1)| = |y + z - 1|\,|y - z|$ has a uniform finite bound on the first factor. Consider any other solution with the same initial value, on the intersection of their intervals. The set where the solutions agree is closed by continuity. At a point of agreement, centre a rectangle there; both graphs remain in it nearby by continuity, so the cited local uniqueness theorem Theorem 33.1 makes the agreement set open. It contains zero; since the intersection is an interval, they agree throughout it [C]. As $t \uparrow \ln 2$, the displayed solution tends to $+\infty$. If another solution of the same initial-value problem were defined on an interval extending beyond $\ln 2$, uniqueness would make it equal to this solution on their overlap. It could then not be finite and continuous at $\ln 2$. Thus $(-\infty, \ln 2)$ is the maximal interval: no solution of this initial problem exists on a larger interval containing it. This conclusion uses the cited uniqueness input [C]. The two equilibria were checked separately and are not obtained by dividing through their zero factors.

33.4 Linear equations and an integrating factor

A scalar *linear first-order equation* has the form $y' + p(t)y = q(t)$, with given coefficients p and q. The equation is *homogeneous* when q is identically zero. Let p and q be continuous on a nonempty open interval I and let $t_0 \in I$. Define the *integrating factor* by

$$P(t) = \int_{t_0}^{t} p(s)\,\mathrm{d}s, \qquad \mu(t) = e^{P(t)} > 0.$$

The fundamental theorem and chain rule give $\mu' = p\mu$. The product rule therefore turns the equation into $(\mu y)' = \mu(y' + py) = \mu q$.

Theorem 33.7 (The scalar linear initial-value formula)**.** *Under these continuity hypotheses, for every $y_0 \in \mathbb{R}$ there is exactly one C^1 solution on I with $y(t_0) = y_0$. It is*

$$y(t) = e^{-P(t)}\left(y_0 + \int_{t_0}^{t} e^{P(s)} q(s)\,\mathrm{d}s\right). \tag{33.1}$$

Proof. The displayed expression is defined on all of I. Every integral has continuous integrand on the finite interval between t_0 and t. The fundamental theorem makes the bracket C^1, and e^{-P} is C^1. Multiplying the formula by μ and differentiating gives $(\mu y)' = \mu q$. Since

$\mu > 0$, the product-rule identity above gives $y' + py = q$. At $t = t_0$, one has $P(t_0) = 0$ and the integral is zero, so the initial value is y_0.

Conversely, any solution satisfies $(\mu y)' = \mu q$. Integration from t_0 to t, with $\mu(t_0) = 1$, forces $\mu(t)y(t) = y_0 + \int_{t_0}^{t} \mu(s)q(s)\,\mathrm{d}s$. Division by the positive factor gives exactly Eq. (33.1). This proves uniqueness on I as well as existence there. ∎

Example 33.8 (A coefficient depending on time). Solve the problem

$$y' + \frac{2t}{1+t^2}y = \frac{t}{1+t^2}, \qquad y(0) = 2.$$

Both coefficients are continuous on $\mathbb{R}$. The factor is $\mu(t) = \exp(\int_0^t 2s/(1+s^2)\,\mathrm{d}s) = 1 + t^2$. Multiplication and integration give

$$((1+t^2)y)' = t, \qquad (1+t^2)y = 2 + \frac{t^2}{2}, \qquad y(t) = \frac{t^2+4}{2(1+t^2)}.$$

The denominator never vanishes, so this is the unique solution on $\mathbb{R}$. It tends to $1/2$ as $t \to \pm\infty$.

33.5 Cooling in a changing environment

Newton's cooling model prescribes the temperature derivative as $T' = -k(T - T_a(t))$, where $k > 0$ and T_a is the given ambient temperature. A negative difference makes the derivative positive: the same model includes heating. Suppose time is measured in minutes, temperature in degrees Celsius, $k = 1/2$ per minute, $T_a(t) = 20 + 2t$ and $T(0) = 60$. The mathematical model is

$$T' + \frac{1}{2}T = 10 + t.$$

Figure 33.2 compares the exact solution with the changing environment.

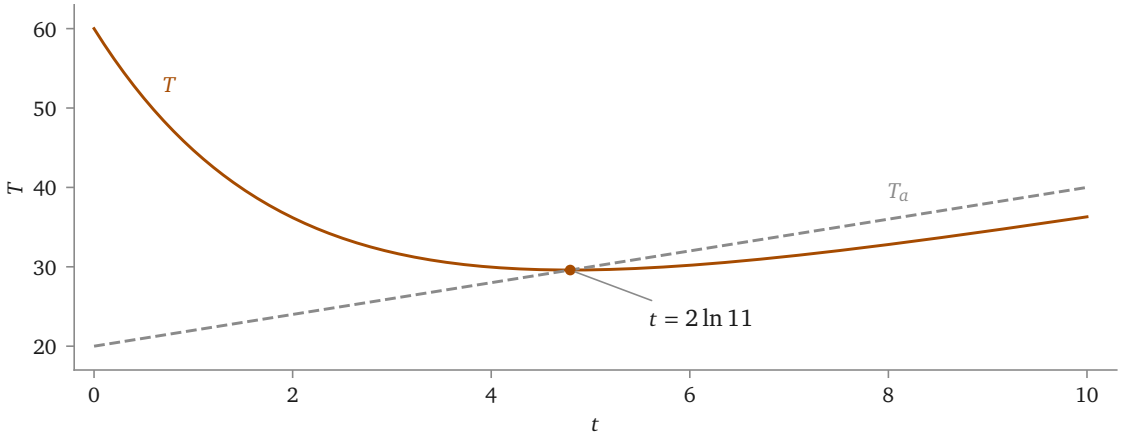


Figure 33.2: The object's temperature starts above the increasing ambient temperature. At the marked intersection its derivative vanishes, after which the object warms while remaining below the ambient temperature.

The factor is $\mu = e^{t/2}$, so $(e^{t/2}T)' = e^{t/2}(10+t)$. Integration by parts evaluates the required primitive: $\int e^{t/2}(10+t)\,\mathrm{d}t = e^{t/2}(2t+16) + C$. The initial-value formula gives

$$T(t) = e^{-t/2}\bigl(60 + e^{t/2}(2t+16) - 16\bigr) = 2t + 16 + 44e^{-t/2}.$$

Substitution verifies $T' + T/2 = 10 + t$ and $T(0) = 60$. The coefficients are continuous for all real time, so the linear theorem makes this the unique mathematical solution on $\mathbb{R}$. The specified future experiment uses $t \geq 0$.

The difference from ambient is $T - T_a = -4 + 44e^{-t/2}$. It decreases through zero once, at $t = 2\ln 11$ minutes. The temperature derivative is $T' = 2 - 22e^{-t/2}$, negative before that time and positive afterwards. The minimum temperature is therefore $20 + 4\ln 11$ degrees Celsius. As $t \to \infty$, the object approaches a temperature 4 degrees below the ambient value, rather than approaching a constant temperature. This conclusion belongs to the stated indefinitely increasing ambient model.

33.6 A complex-valued first-order equation

Let $\lambda, z_0 \in \mathbb{C}$, and let t remain real. Lecture 16 defines complex derivatives through their real and imaginary components and proves the product rule and $(e^{\lambda t})' = \lambda e^{\lambda t}$. These facts give a uniqueness theorem without using Picard–Lindelöf.

Theorem 33.9 (Complex exponential initial-value problem). *On every open interval I containing t_0, the unique differentiable complex-valued solution of $z' = \lambda z$, $z(t_0) = z_0$, is*

$$z(t) = z_0 e^{\lambda (t - t_0)}.$$

Proof. Differentiation verifies the equation for the displayed function and its value at t_0 is z_0. For any solution, the product rule gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(e^{-\lambda (t-t_0)} z(t)\right) = e^{-\lambda (t-t_0)}(z' - \lambda z) = 0.$$

Both real components of this product have zero derivative, hence each is constant on I by Lecture 9. Evaluating the product at t_0 makes its constant value z_0. The complex exponential is nonzero and its inverse is $e^{\lambda (t-t_0)}$, so multiplication gives the asserted formula. The proof includes $\lambda = 0$ and $z_0 = 0$. ∎

For example, the data $z' = (-1 + 2\mathrm{i})z$, $z(0) = 1 + \mathrm{i}$, give

$$z(t) = e^{-t}\big((\cos 2t - \sin 2t) + \mathrm{i}(\sin 2t + \cos 2t)\big).$$

Its modulus is $\sqrt{2}e^{-t}$ and its angle advances at 2 radians per unit time. These are real-parameter statements. No derivative with respect to a complex increment has been introduced.

Summary 33.

- A solution satisfies its equation at every point of its stated interval. Theorem 33.1 gives local existence and uniqueness under continuity and a uniform local Lipschitz bound in the unknown value [C]. It does not assert global existence.
- Theorem 33.4 justifies separation on value intervals where B is nonzero. Equilibria $B(\eta) = 0$ are checked before division.
- For continuous p and q on an open interval, the positive factor $e^{\int_{t_0}^{t} p}$ gives the unique linear solution Eq. (33.1) on that interval, for every real initial value.
- Theorem 33.9 proves $z(t) = z_0 e^{\lambda (t-t_0)}$ uniquely for complex constants and real time. Its proof uses the zero-derivative theorem componentwise.

Sheet 33 develops nonlinear models and linear applications, retaining all division and interval guards. Lecture 34 uses the complex uniqueness result to prove that the characteristic-root list for a second-order equation is complete.

References

G. Teschl, *Ordinary Differential Equations and Dynamical Systems*, Graduate Studies in Mathematics 140, AMS, 2012, §2.2, Theorem 2.2, p. 38 (Picard–Lindelöf [C]); author's preliminary version. J. Stewart, D. Clegg and S. Watson, *Calculus: Early Transcendentals*, 9th ed., Cengage, 2020, §9.3 and §9.5.